



BLUETEAMERS

—DEFEND. DETECT. RESPOND.—

MITRE ATT&CK EXPLAINED FOR SOC ANALYSTS



WHAT IT IS • WHY IT MATTERS • HOW TO USE IT
IN REAL SOC INVESTIGATIONS



UNDERSTAND
ATTACKER
BEHAVIOR



IMPROVE
DETECTION &
INVESTIGATION



ENHANCE
THREAT
HUNTING



STRENGTHEN
TEAM
COMMUNICATION



BUILD BETTER
DEFENSES



MITRE ATT&CK helps SOC Analysts understand attacker tactics and techniques, map alerts to known behaviors, investigate incidents systematically and respond with confidence.



Map.
Detect.
Investigate.
Respond.



PREPARED BY BLUETEAMERS



blueteamers_



blueteamers.io



WHAT IS MITRE ATT&CK?

MITRE ATT&CK is a public knowledge base that documents real-world adversary behavior and tactics.

It provides a common language for the cybersecurity community to describe, understand and defend against attacks.



IN SIMPLE WORDS



MITRE ATT&CK is like a playbook of how attackers think and what actions they take during an attack.

It does not focus on tools or malware.

It focuses on **techniques** and **behaviors**.

“ ATT&CK is not a product.
It's a knowledge base.
It's not about how attackers do it, it's about what they do.

”

HOW IT WORKS



ATTACKER

Performs an action during an attack



MITRE TECHNIQUE

The action is mapped to a specific technique in ATT&CK



SOC ALERT

Your security tools generate an alert based on evidence



INVESTIGATION

Analyst investigates using the context of the technique



RESPONSE

Incident is handled, contained and lessons are learned

KEY TAKEAWAYS



- ✓ MITRE ATT&CK is a global knowledge base of adversary tactics and techniques.
- ✓ It helps defenders understand how attacks happen in the real world.
- ✓ It standardizes the language between tools, teams and organizations.
- ✓ It improves detection, investigation, threat hunting and incident response.
- ✓ It is used by SOC teams, threat hunters, incident responders and red teams.





WHY WAS MITRE ATT&CK CREATED?

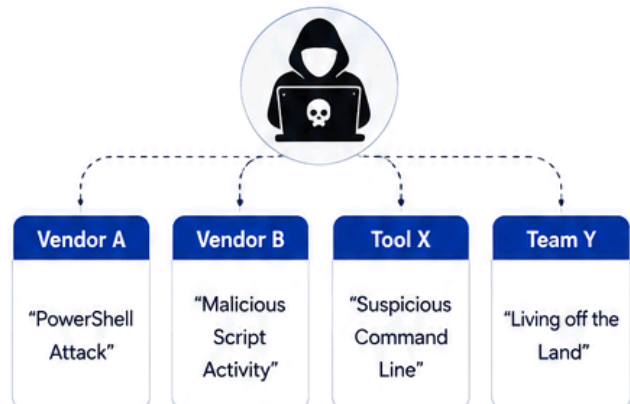
Before MITRE ATT&CK, every security vendor, tool and team used different names to describe the same attacker behavior.

This created confusion, made threat intelligence hard to share and investigations inconsistent.



MITRE ATT&CK was created to standardize the language around adversary behavior.

THE PROBLEM BEFORE MITRE ATT&CK



Different names. Same behavior.
Difficult to detect, investigate and respond.

AFTER MITRE ATT&CK

Everyone now speaks the same language.



Behavior Observed
PowerShell used by attacker



Mapped to ATT&CK
Execution via PowerShell



MITRE Technique
T1059.001 – PowerShell

EXAMPLE COMPARISON

BEFORE (Inconsistent)	AFTER (Standardized)
"PowerShell Attack"	Execution – T1059
"Suspicious Script Activity"	T1059.001 – PowerShell
"Malicious Command Line"	Execution – Command and Scripting Interpreter
"Living off the Land"	Defense Evasion / Execution Techniques



Now, analysts, tools, threats feeds and vendors are aligned using a common framework.

WHY THIS MATTERS FOR SOC TEAMS



COMMON LANGUAGE

Everyone uses the same names and IDs to talk about attacker behavior.



BETTER COLLABORATION

Easier to share threat intelligence and align across teams and tools.



IMPROVED DETECTION

Map detections to known techniques and reduce blind spots.



FASTER INVESTIGATION

Understand attacker intent quickly and follow a structured approach.



STRONGER DEFENSES

Build detections, detections, hunting queries and controls based on real attacker behavior.



Key Takeaway: MITRE ATT&CK was created to bring clarity, consistency and structure to how we understand and defend against real-world attacks.



WHY DO SOC ANALYSTS USE MITRE ATT&CK?

MITRE ATT&CK helps SOC Analysts understand what attackers are trying to do, not just what tools they use.

It provides context to alerts, connects the dots between events and helps drive better investigations, detections and responses.

THE BIG PICTURE

MITRE ATT&CK bridges the gap between attacker actions and defender visibility.



It helps analysts focus on the "why" behind an alert, not just the "what".

HOW SOC ANALYSTS USE MITRE ATT&CK EVERY DAY



MITRE ATT&CK turns raw alerts into meaningful insights and stronger security outcomes.

REAL-WORLD EXAMPLE



By mapping each step to MITRE ATT&CK, you can understand the attacker's full plan and respond effectively.

KEY BENEFITS



FOCUS ON ATTACKER BEHAVIOR

Understand what attackers do, not just what malware or tools they use.



COMMON LANGUAGE

Use the same framework across teams, vendors and tools.



BETTER DETECTIONS

Build detections based on techniques attackers actually use.



FASTER INVESTIGATIONS

Leverage ATT&CK data to guide and validate your investigation path.



STRONGER DEFENSES

Hunt threats proactively and measure coverage against real attacker behavior.



PRO TIP: Don't just stop at the alert. Map it. Understand it. Hunt it. Mitigate it. MITRE ATT&CK is your roadmap to **think like an attacker** and **defend like a pro**.

10101
01010





MITRE ATT&CK MATRIX EXPLAINED

The MITRE ATT&CK Matrix is a representational model of the tactics and techniques that adversaries use during an attack.

It helps defenders organize, understand and map attacker behavior in a structured way.



Think of the Matrix as a map of how an attack happens from start to finish.



Each column = a **Tactic** | Each item = a **Technique**
(techniques may have sub-techniques)

ENTERPRISE ATT&CK MATRIX (TACTICS)

Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	...	Impact
10 techniques	10 techniques	17 techniques	14 techniques	43 techniques	...	13 techniques
- T1566.001 - T1078.004 - T1190 - T1133 - T1033 ...	- T1059.001 - T1059.003 - T1204.002 - T1106 - T1021.002 ...	- T1053.005 - T1547.001 - T1543.003 - T1136.001 - T1505.003 ...	- T1068 - T1548.002 - T1574.002 - T1134 - T1055.001 ...	- T1027 - T1055 - T1562.001 - T1036 - T1070.004 ...	...	- T1486 - T1489 - T1490 - T1491 - T1531 ...

KEY COMPONENTS



TACTIC

Represents the adversary's short-term goal or objective during an attack. There are 14 tactics in the Enterprise ATT&CK framework.

Example:

Initial Access, Execution, Persistence, Credential Access, Discovery, etc.



TECHNIQUE

Represents how the adversary achieves a tactical goal. There are hundreds of techniques.

Example:

PowerShell (T1059), Phishing (T1566), Scheduled Task (T1053)



SUB-TECHNIQUE

A more detailed variant of a technique that represents a specific method.

Example:

PowerShell (T1059.001), Spearphishing Attachment (T1566.001)

EXAMPLE WALKTHROUGH

		TACTIC	TECHNIQUE	DESCRIPTION
	Phishing email received →	Initial Access	Phishing (T1566.001)	Attacker gained access via spearphishing email attachment.
	PowerShell executed →	Execution	PowerShell (T1059.001)	Attacker executed PowerShell using encoded command.
	Scheduled task created for persistence →	Persistence	Scheduled Task (T1053.005)	Created a scheduled task to maintain persistence.
	Credentials dumped →	Credential Access	Credential Dumping (T1003.001)	Dumped credentials from LSASS memory.
	Data exfiltrated to cloud →	Exfiltration	Exfiltration Over Web Service (T1567.002)	Stole data and exfiltrated it to external cloud storage.



Key Takeaway: The MITRE ATT&CK Matrix helps SOC Analysts see the bigger picture, understand attacker intent and follow a structured approach to investigate and respond.





THE 14 ENTERPRISE TACTICS

MITRE ATT&CK organizes attacker behavior into 14 tactics that represent the "why" behind an action.



Tactics describe the adversary's short-term goal or objective.

AN ATTACK IS A JOURNEY



Attacker



Access



Execute



Exploit



Achieve Objective

The 14 tactics show how attackers move from initial access to final impact.

THE 14 ENTERPRISE TACTICS AT A GLANCE



RECONNAISSANCE

Adversaries gather information about the target.



RESOURCE DEVELOPMENT

Adversaries develop resources to support their operations.



INITIAL ACCESS

Adversaries gain access to the victim environment.



EXECUTION

Adversaries run malicious code or commands.



PERSISTENCE

Adversaries maintain access to the environment.



PRIVILEGE ESCALATION

Adversaries gain higher-level permissions.



DEFENSE EVASION

Adversaries evade detections and security controls.



CREDENTIAL ACCESS

Adversaries steal or discover valid credentials.



DISCOVERY

Adversaries discover systems, users and network details.



LATERAL MOVEMENT

Adversaries move within the network to reach more systems.



COLLECTION

Adversaries collect data from the environment.



COMMAND AND CONTROL

Adversaries communicate with controlled external infrastructure.



EXFILTRATION

Adversaries steal data from the environment.



IMPACT

Adversaries achieve their goal and cause harm.

TACTIC VS TECHNIQUE



TACTIC (The Why)

Represents the adversary's goal or objective.



TECHNIQUE (The How)

Represents how the adversary achieves that goal.

Example:

Tactic: Execution (Why)

Technique: PowerShell (How)

ID: T1059.001

HOW TACTICS HELP SOC ANALYSTS



Provide context to alerts and suspicious activities.



Help connect multiple events into a clear attack story.



Guide investigations in the right direction.



Improve threat hunting by focusing on attacker goals.



Strengthen detections and response strategies.

KEY TAKEAWAY



Understanding the 14 tactics helps you think like an attacker and defend like a pro.



Pro Tip: Always ask yourself – "What is the attacker trying to achieve at this stage?" Identifying the tactic helps you choose the right investigation path.





TECHNIQUES & SUB-TECHNIQUES EXPLAINED

MITRE ATT&CK breaks down attacker behavior into Techniques and Sub-techniques.

This helps SOC Analysts understand exactly what attackers are trying to do and how they do it.



Mapping alerts to the right technique helps you investigate **faster** and more **accurately**.

TECHNIQUE HIERARCHY



TACTIC (The Why)

The adversary's short-term goal or objective.

Example: Execution



TECHNIQUE (The How)

The adversary's tactical goal that achieves the objective.

Example: Command and Scripting Interpreter (T1059)



SUB-TECHNIQUE (The Detail)

A specific method or variant used to achieve the technique.

Example: PowerShell (T1059.001)



Sub-techniques give more granularity and help you detect attacker behavior with precision.

REAL-WORLD EXAMPLES



POWERHELL

Attackers abuse PowerShell to run commands, download payloads or perform malicious activities.

Technique:

Command and Scripting Interpreter

T1059.001



COMMAND PROMPT

Attackers use CMD to execute commands and carry out system manipulation.

Technique:

Command and Scripting Interpreter

T1059.003



SCHEDULED TASK

Attackers create or modify scheduled tasks to maintain persistence on the system.

Technique:

Scheduled Task/Job

T1053.005



SPEARPHISHING ATTACHMENT

Attackers send malicious attachments to trick users and gain initial access.

Technique:

Phishing Attachment

T1566.001



VALID ACCOUNTS

Attackers use legitimate accounts to blend in and avoid detection.

Technique:

Valid Accounts

T1078

WHY THIS MATTERS FOR SOC ANALYSTS



Helps you understand the attacker's **intent** not just the tool used.



Provides **context** to alerts and reduces false positives.



Guides you to **collect the right logs** and evidence.



Improves **detection engineering** and **threat hunting**.



Ensures **consistent communication** across tools and teams.

KEY TAKEAWAY



Techniques tell you **how** the attacker achieves a goal.



Sub-techniques tell you the **exact** method they use.



Together, they help you investigate **better**, detect **smarter** and respond **faster**.





HOW A REAL ATTACK MAPS TO MITRE ATT&CK

Attackers follow steps to achieve their goal.
MITRE ATT&CK helps us map every step
to a Tactic and Technique.

WHAT THIS SHOWS



Shows how attackers move through different tactics.



Helps defenders understand where to detect and respond.



Provides a complete attack story with MITRE IDs for every step.

ATTACK FLOW EXAMPLE

	ATTACK STEP	TACTIC	TECHNIQUE	MITRE ID	DESCRIPTION
1	 Phishing Email Received	Initial Access	Spearphishing Link / Attachment	T1566.001	Attacker sends a phishing email with a malicious attachment.
2	 User Opens Attachment	Execution	User Execution: Malicious File	T1204.002	User opens the file, triggering malicious code execution.
3	 PowerShell Executes	Execution	Command and Scripting Interpreter: PowerShell	T1059.001	Malicious PowerShell command retrieves and executes payload.
4	 Scheduled Task Created	Persistence	Scheduled Task / Job	T1053.005	Attacker creates a scheduled task to maintain persistence.
5	 Credential Dumping	Credential Access	Credential Dumping: LSASS Memory	T1003.001	Attacker dumps credentials from LSASS to escalate their access.
6	 Beacons to C2 Server	Command & Control	Application Layer Protocol	T1071.001	Malware communicates with C2 server over HTTPS.
7	 Data Exfiltration	Exfiltration	Exfiltration Over Web Service	T1041	Attacker steals sensitive data and exfiltrates it to external server.

THE FULL ATTACK CHAIN



KEY TAKEAWAYS



- ✓ Attackers follow a path – you can map it.
- ✓ MITRE ATT&CK gives every step a name and ID.
- ✓ Mapping helps you detect early and respond faster.
- ✓ Knowing the next possible step helps in threat hunting.



PRO TIP FOR SOC ANALYSTS

Always ask:

- What tactic is the attacker in?
- What technique are they using?
- What could be their next move?

This mindset turns alerts into insights and insights into action.



Remember: Attackers may skip steps or move out of order.
But their actions will always leave evidence – **map it with MITRE!**





HOW SOC ANALYSTS USE MITRE ATT&CK EVERY DAY

MITRE ATT&CK is not just a framework, it is part of the daily workflow of a SOC Analyst.

It helps us investigate, detect, respond and improve security.

“

We don't just stop at the alert. We map it, investigate it, understand it and stop it.

SOC ANALYST WORKFLOW



1 ALERT GENERATED

SIEM, EDR or other tools generate an alert based on suspicious activity.



2 IDENTIFY TECHNIQUE

Analyst maps the alert to a MITRE ATT&CK Technique to understand attacker behavior.



3 COLLECT LOGS & EVIDENCE

Collect relevant logs from endpoints, network, cloud, email, firewall, etc.



4 VALIDATE ACTIVITY

Analyze and validate if the activity is malicious and understand the full context.



5 CONTAIN & ERADICATE

Take action to contain the threat and remove attacker's access.



6 ESCALATE & REPORT

Escalate if required and document the incident with MITRE mapping for reporting.

COMMON LOG SOURCES ANALYSTS USE



Sysmon Logs

Process creation, network connections, file events and more.



Windows Event Logs

Security, System, Application and PowerShell logs.



Microsoft Defender Logs

Device alerts, malware detections and behavior logs.



Firewall Logs

Track allowed, blocked connections and suspicious traffic.



DNS Logs

Identify suspicious domains and DNS tunneling.



EDR Logs

Process behavior, file reputation, threat detections.



Proxy Logs

Web requests, downloads and user activity.



SIEM Logs

Centralized logs, correlation and alert generation.

HOW MITRE HELPS AT EACH STAGE



Identify

MITRE provides context to understand what the attacker is trying to do.



Investigate

Helps find related events and build the full attack story.



Detect

Use MITRE techniques to create better detections and reduce false positives.



Respond

Know the attacker's goal and take the right action to stop them.



Improve

Continuously improve detections and threat hunting using MITRE data.



MITRE ATT&CK is the common language that helps SOC teams think like attackers and defend like pros.

ANALYST MINDSET



THINK LIKE AN ATTACKER

Use MITRE to understand attacker intent and possible next steps.



LOOK FOR PATTERNS

Map alerts to techniques and find patterns across systems.



CONNECT THE DOTS

Build the full attack chain using MITRE techniques.



FOCUS ON OUTCOMES

Understand the attacker's objective and prevent it.



KEY TAKEAWAY: MITRE ATT&CK guides us in every step of an investigation. From the first alert to the final report.





HOW SIEM MAPS ALERTS TO MITRE ATT&CK

SIEM and security tools use MITRE ATT&CK to classify, organize and help analysts investigate alerts consistently.

WHY THIS MATTERS



Provides a common language across tools and security teams.



Helps analysts understand attacker behavior and speed up investigations.



Improves detection engineering and threat hunting.

ALERT TO MITRE MAPPING FLOW

1. LOGS INGESTED



Logs are collected from endpoints, network, cloud, identity, etc.

2. SIEM PROCESSING



SIEM parses and normalizes the logs.

3. DETECTION RULE TRIGGERED



A correlation rule or detection rule detects suspicious activity.

4. MITRE MAPPING



The alert is mapped to a MITRE ATT&CK Technique and Tactic.

5. SOC ALERT



Analyst receives an alert with context and MITRE mapping.

“ MITRE mapping turns raw alerts into meaningful threat intelligence.

HOW DIFFERENT TOOLS MAP ALERTS TO MITRE



Security Alert

Rule Name	Suspicious PowerShell
Severity	High
MITRE Tactic	Execution
MITRE Technique	T1059.001

Details

HOW IT WORKS

Splunk ES uses correlation searches and lookups to map alerts to MITRE techniques.

EXAMPLE MAPPING

T1059.001

PowerShell
(Execution)



Security Alert

Incident ID	12345
Severity	High
Tactic	Execution
Technique	T1059.001
Description	PowerShell activity detected

HOW IT WORKS

Sentinel uses built-in analytics rules that are mapped to MITRE ATT&CK.

EXAMPLE MAPPING

T1059.001

PowerShell
(Execution)



Alert Details

Rule ID	100002
Level	12
Tactic	Execution
Technique	T1059.001
Description	PowerShell command detected

HOW IT WORKS

Wazuh rules are mapped to MITRE techniques using rule metadata and decoders.

EXAMPLE MAPPING

T1059.001

PowerShell
(Execution)



Offense

Offense ID	67890
Magnitude	7
Tactic	Execution
Technique	T1059.001
Description	Possible PowerShell execution

HOW IT WORKS

QRadar offenses include MITRE mapping for better context and prioritization.

EXAMPLE MAPPING

T1059.001

PowerShell
(Execution)



Alert

Risk Score	73
Tactic	Execution
Technique	T1059.001
Rule	Suspicious PowerShell Activity

HOW IT WORKS

Elastic Security rules are tagged with MITRE ATT&CK tactics and techniques.

EXAMPLE MAPPING

T1059.001

PowerShell
(Execution)



Different tools, Same Framework – MITRE ATT&CK.

No matter which tool you use, the mapping helps you understand the attacker's behavior.



REAL EXAMPLE: POWERSHELL DETECTION

Process: powershell.exe
Command Line: powershell.exe -enc SQBFAFgA...
Parent Process: winword.exe
User: john.doe
Severity: High



MITRE MAPPING

Tactic: Execution
Technique: Command and Scripting Interpreter
ID: T1059.001
Sub-technique: PowerShell

This mapping gives context and helps analysts investigate faster.

BENEFITS OF SIEM MITRE MAPPING

- ✓ Standardizes alert classification
- ✓ Improves investigation speed
- ✓ Enhances threat hunting
- ✓ Enables better reporting
- ✓ Strengthens communication between teams
- ✓ Supports compliance and audit requirements



Key Takeaway: MITRE mapping in SIEM tools turns alerts into actionable insights, helps you understand attacker behavior and improves overall SOC effectiveness.





MITRE ATT&CK VS CYBER KILL CHAIN

Both frameworks help us understand attacks, but they serve different purposes and work at different levels.

THE BIG PICTURE



MITRE ATT&CK

Focuses on **what** attackers do.
Detailed and granular.



CYBER KILL CHAIN

Focuses on the **phases** of an attack.
High-level and linear.

“

Use both together to prevent, detect and respond more effectively.

FEATURE	MITRE ATT&CK	VS	CYBER KILL CHAIN
FOCUS	Focuses on attacker behavior (Techniques and Tactics)		Focuses on attack lifecycle (Phases of an attack)
STRUCTURE	14 Tactics, many Techniques and Sub-techniques		7 sequential stages
LEVEL OF DETAIL	Very detailed and granular		High-level and broad
PURPOSE	Understand and categorize adversary behavior		Understand the overall attack lifecycle
USE CASE	Detection, Threat Hunting, Incident Response, Purple Teaming, Threat Intelligence		Attack Modeling, Awareness, Strategy, High-level Training
AUDIENCE	SOC Analysts, Engineers, Threat Hunters, IR Teams		Executives, Beginners, Non-technical Teams
FLEXIBILITY	Non-linear, adversaries can skip steps or go back		Linear and sequential
DATA SOURCE	Logs, Alerts, Threat Intel, Telemetry, EDR, SIEM, etc.		Intelligence, Reports, Attack Observations
EXAMPLE	T1059 (Command and Scripting Interpreter)		Delivery → Exploitation → Installation → C2 → Actions on Objectives
BEST FOR	Answering “What is the attacker doing right now?”		Answering “Where is the attacker in the attack lifecycle?”

CYBER KILL CHAIN 7 STAGES

- RECONNAISSANCE**
Attacker researches the target.
- WEAPONIZATION**
Attacker creates a malicious payload.
- DELIVERY**
Payload delivered to the target.
- EXPLOITATION**
Attacker exploits a vulnerability.
- INSTALLATION**
Malware is installed on the system.
- COMMAND & CONTROL**
Attacker establishes C2 channel.
- ACTIONS ON OBJECTIVES**
Attacker achieves their goals.

WHEN SHOULD YOU USE WHICH?

USE MITRE ATT&CK WHEN YOU:

- ✓ Investigate alerts and incidents
- ✓ Hunt threats in your environment
- ✓ Build detections and SIEM rules
- ✓ Map adversary behavior
- ✓ Work with EDR, Threat Intel or Purple Teaming

USE CYBER KILL CHAIN WHEN YOU:

- ✓ Model attacks at a high level
- ✓ Create security awareness content
- ✓ Explain attacks to non-technical teams
- ✓ Build security strategies and roadmaps
- ✓ Understand attacker lifecycle from start to finish

KEY TAKEAWAY: MITRE ATT&CK gives you the details.
Cyber Kill Chain gives you the big picture.
Together, they make you a better defender!

“

Think detailed with MITRE.
Think strategic with Kill Chain.
Defend with both.





TOP 15 MITRE ATT&CK TECHNIQUES EVERY SOC ANALYST SHOULD KNOW

These techniques are commonly used by attackers. Knowing them helps you detect, investigate and respond faster.

WHY THESE MATTER



These techniques appear in real-world attacks every day.



They are used across multiple tactics and attack chains.



Focusing on these will improve your detection and investigation skills.

THE 15 MUST-KNOW TECHNIQUES

1



PowerShell
Execution

T1059.001

Attackers use PowerShell to run commands and execute malicious code.

2



Phishing
Initial Access

T1566.001

Attackers send phishing emails to trick users into malicious actions.

3



Credential Dumping
Credential Access

T1003.001

Attackers extract credentials from memory or credential stores.

4



Scheduled Task
Persistence

T1053.005

Attackers create scheduled tasks to maintain access over time.

5



Valid Accounts
Initial Access

T1078

Attackers use legitimate accounts to access systems and move laterally.

6



Remote Services
Lateral Movement

T1021.002

Attackers use remote services to move between systems.

7



Brute Force
Credential Access

T1110

Attackers try multiple passwords to gain unauthorized access.

8



Ingress Tool Transfer
Command & Control

T1105

Attackers transfer tools onto the system to aid their attack.

9



Exfiltration Over Web Service
Exfiltration

T1041

Attackers exfiltrate data over web protocols like HTTPS.

10



Application Layer Protocol
Command & Control

T1071.001

Attackers use application layer protocols to establish C2 communication.

11



Boot or Logon Autostart Execution
Persistence

T1547.001

Attackers configure systems to run at startup or user logon.

12



Signed Binary Proxy Execution
Defense Evasion

T1218

Attackers abuse legitimate binaries to execute malicious payloads.

13



System Information Discovery
Discovery

T1082

Attackers gather information about the system, OS, network, and software.

14



Remote System Discovery
Discovery

T1018

Attackers scan and identify other systems in the network.

15



File and Directory Discovery
Discovery

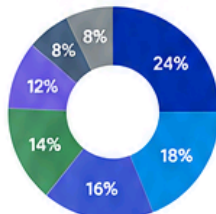
T1083

Attackers look for files and folders of interest on the compromised system.

QUICK TIPS

- ✓ Memorize IDs of common techniques.
- ✓ Understand what attackers are trying to do.
- ✓ Map alerts to these techniques.
- ✓ Use MITRE to guide your investigation.
- ✓ Always hunt for these in your environment.

MOST COMMON TACTICS USED



- Execution (24%)
- Credential Access (18%)
- Persistence (16%)
- Initial Access (14%)
- Command & Control (12%)
- Discovery (8%)
- Exfiltration (8%)



Focus on understanding attacker behavior, not just memorizing IDs.

The more you practice mapping alerts to these techniques, the better you'll become!



KEY TAKEAWAY: These 15 techniques appear in most real-world attacks. Master them and you'll be able to detect, investigate and respond like a pro!





INVESTIGATING A REAL SECURITY ALERT WITH MITRE ATT&CK

“Don't just look at the alert. Understand the attacker behavior behind it.

MITRE helps you see the full picture.

SCENARIO: Microsoft Defender generates a PowerShell alert. Instead of panicking, the analyst uses MITRE to understand what happened, why it happened and what to do next.

SOC INVESTIGATION FLOW

- 1 ALERT GENERATED**
Microsoft Defender detects suspicious PowerShell activity and generates an alert.
- 2 IDENTIFY MITRE TECHNIQUE**
Map the alert to MITRE ATT&CK.
Technique: **T1059.001 (PowerShell)**
- 3 CHECK PARENT PROCESS**
Identify the parent process that launched PowerShell.
- 4 REVIEW COMMAND LINE**
Analyze the PowerShell command line for suspicious patterns or encoded data.
- 5 CHECK NETWORK CONNECTIONS**
Look for outbound connections, downloads or C2 communication.
- 6 THREAT INTELLIGENCE LOOKUP**
Search hashes, IPs, domains and URLs in threat intel sources.
- 7 CONTAIN**
Isolate the endpoint, block IOCs and stop attacker from moving further.
- 8 ESCALATE**
Document findings, escalate to L2/L3 and update the case.



Why this matters?

Following a structured process with MITRE helps you investigate faster and with confidence.

TOOLS YOU MIGHT USE



POWERSHELL ALERT EXAMPLE



ALERT

Suspicious PowerShell Activity Detected on Endpoint: WIN-7G26B1



PROCESS

powershell.exe



PARENT PROCESS

winword.exe
(User opened a malicious document)



COMMAND LINE (ENCODED)

powershell.exe -enc SQBFAGa...
(Encoded PowerShell command)



BEHAVIOR

Downloads payload from:
[hxxps://malicious-site\[.\]com/payload.ps1](https://malicious-site[.]com/payload.ps1)



NETWORK ACTIVITY

Outbound HTTPS connection to 203.0.113.45:443



MITRE MAPPING

Tactic: Execution

Technique: Command and Scripting Interpreter

T1059.001 (PowerShell)

WHAT WE LEARNED

- ✓ Attacker used a malicious document to run PowerShell.
- ✓ Used encoded command to hide intent.
- ✓ Downloaded payload from external server.
- ✓ Technique mapped to T1059.001 (Execution).
- ✓ Further actions may include persistence or C2.



KEY TAKEAWAY MITRE ATT&CK gives you context and clarity.

It turns a simple alert into a meaningful investigation and helps you follow the right steps.





MITRE ATT&CK IN THREAT HUNTING

Threat hunting is the proactive search for malicious activity that may not be detected by security alerts.

“ Instead of waiting for alerts, hunt for attacker behaviors using MITRE ATT&CK.”

WHAT IS THREAT HUNTING?



Threat Hunting is a proactive approach where analysts search for attacker activity that has bypassed detections. MITRE ATT&CK provides the map to understand what attackers do, so we can hunt for them in our environment.

IOC vs TTP

IOC (Indicators of Compromise)

- Known bad indicators (IP, Hash, Domain)
- Good for detecting known threats

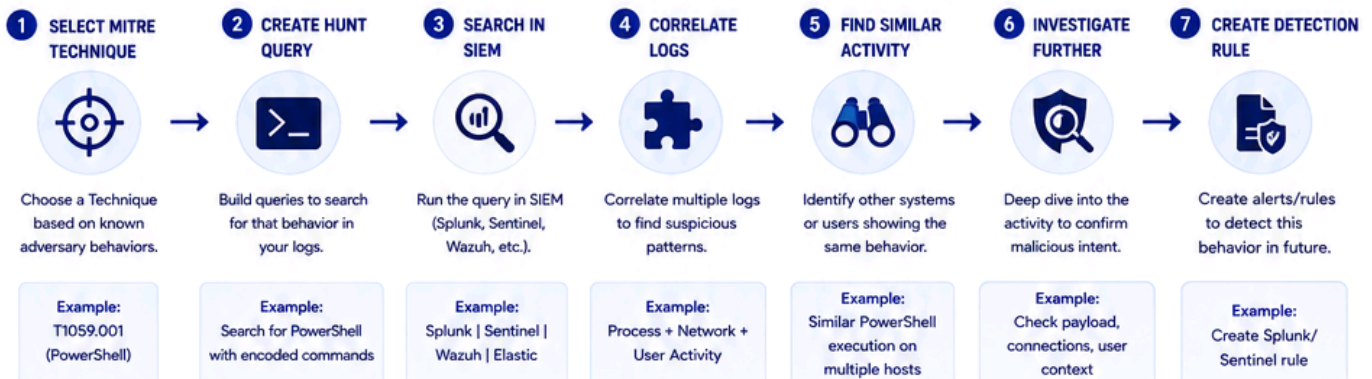
TTP (Tactics, Techniques, Procedures)

- How attackers perform actions (PowerShell execution, Credential Dumping)
- Good for finding unknown threats

WHY MITRE IMPROVES HUNTING

- ✓ Focuses on attacker behavior not just indicators
- ✓ Provides a structured way to search for threats
- ✓ Helps uncover advanced persistent threats (APTs)
- ✓ Improves detection coverage and reduces blind spots

THREAT HUNTING WORKFLOW WITH MITRE ATT&CK



EXAMPLE HUNT: POWERSHELL EXECUTION (T1059.001)



MITRE Technique
T1059.001 – PowerShell



Goal
Find suspicious PowerShell execution that may indicate malicious activity.



Hunt Query Example (Splunk)

```
index=*>ineventlog EventCode=4688
New_Process_Name="*\powershell.exe"
| search CommandLine="**enc** OR CommandLine="**e**"
| table _time, ComputerName, User, CommandLine, Parent_Process_Name
```



What It Finds
Encoded PowerShell commands often used to hide malicious scripts.



Next Step
Investigate parent process, command line content, network connections and user context.



HUNT SOURCES (WHERE TO LOOK)

	Windows Logs	Security, System, Application, PowerShell, 4688, 4624, etc.
	Sysmon Logs	Detailed process creation, network connections, etc.
	Microsoft Defender	Alerts, DeviceEvents, Advanced Hunting data.
	Firewall Logs	Allowed/blocked connections, external IPs, ports.
	DNS Logs	Suspicious domains, DNS tunneling.
	Proxy Logs	Web requests, downloads, file transfers.
	EDR Logs	Process behavior, memory analysis, script execution.

TIPS FOR EFFECTIVE HUNTING



Focus on TTPs not just IOCs.



Use MITRE ATT&CK as your hunting guide.



Hunt regularly, not just during incidents.



Collaborate with threat intel and other teams.



Turn your hunts into detections.



KEY TAKEAWAY

MITRE ATT&CK gives you visibility into attacker behaviors. Use it to hunt proactively, find what others miss, and stay one step ahead of attackers.





MOST ASKED MITRE ATT&CK INTERVIEW QUESTIONS

These questions are commonly asked in SOC Analyst interviews. Use them to test your knowledge and improve your confidence.

Q1

WHAT IS MITRE ATT&CK?

MITRE ATT&CK is a knowledge base of adversary tactics and techniques based on real-world observations. It helps organizations understand attacker behavior and improve their defenses.



Q2

DIFFERENCE BETWEEN TACTIC, TECHNIQUE AND SUB-TECHNIQUE?

- **Tactic:** The "why" (attacker's goal)
- **Technique:** The "how" (attacker's action)
- **Sub-technique:** A more specific implementation of a technique.



Q3

EXPLAIN MITRE TECHNIQUE T1059.

T1059 is Command and Scripting Interpreter. It covers attacker techniques where they execute commands or scripts using interpreters like PowerShell, Bash, or CMD.



Q4

WHY DO SOC ANALYSTS USE MITRE ATT&CK?

- Standardizes threat knowledge
- Improves detection engineering
- Helps in threat hunting
- Supports incident response
- Improves communication across teams



Q5

DIFFERENCE BETWEEN MITRE ATT&CK AND CYBER KILL CHAIN?

MITRE ATT&CK is more detailed and flexible with 14 tactics and hundreds of techniques. Cyber Kill Chain has 7 stages and is more linear and high-level.



Q6

HOW DOES SPLUNK USE MITRE ATT&CK?

Splunk maps detections, alerts and security content to MITRE techniques. It helps analysts understand what attacker behavior an alert represents and speeds up investigations.



Q7

GIVE ONE EXAMPLE OF AN ATTACK MAPPED TO MITRE ATT&CK.

Phishing email with a malicious attachment → T1566.001 (Phishing: Spearphishing Attachment) which leads to PowerShell execution → T1059.001 (PowerShell).



Q8

HOW DOES MITRE ATT&CK HELP IN THREAT HUNTING?

- Helps identify attacker TTPs to look for
- Guides creation of hunt queries
- Improves coverage of hidden threats
- Helps prioritize high-risk techniques



BONUS INTERVIEW QUESTIONS

- ▶ Q9 What are the 14 tactics in MITRE ATT&CK?
- ▶ Q10 How do you map an alert to MITRE ATT&CK?
- ▶ Q11 What is the purpose of sub-techniques?
- ▶ Q12 Where can you find MITRE ATT&CK data?

- ▶ Q13 What is Tactic vs Technique mapping in SIEM?
- ▶ Q14 How does MITRE help in Incident Response?
- ▶ Q15 Can MITRE ATT&CK be used for malware analysis?



INTERVIEW TIP

Don't just memorize technique IDs.
Explain how the technique appears in a real-world scenario and how you would investigate it.



KEY TAKEAWAY

The interviewer wants to know how you think.
Use MITRE to explain attacker behavior, not just to answer questions.





MITRE ATT&CK CHEAT SHEET

Your quick reference guide to understand, detect and respond to attacker behavior.

MOST IMPORTANT TECHNIQUES (For SOC)

**T1059**

Command & Scripting
Interpreter
Execution

**T1566**

Phishing
Initial Access

**T1003**

Credential Dumping
Credential Access

**T1053**

Scheduled Task
Persistence

**T1078**

Valid Accounts
Initial Access

**T1041**

Exfiltration Over
C2 Channel
Exfiltration

**T1071**

Application Layer
Protocol
Command & Control

**T1110**

Brute Force
Credential Access

THE MITRE ATT&CK FRAMEWORK



1. 14 TACTICS (GOALS)

Represent the attacker's goals in an attack lifecycle.



2. TECHNIQUES (ACTIONS)

Describe how adversaries achieve their goals.



3. SUB-TECHNIQUES (DETAILS)

Provide more granular details of how techniques are performed.



4. DETECTION

Map techniques to data sources and create detections.



5. INVESTIGATION

Use MITRE to guide investigations and understand attacker behavior.



6. THREAT HUNTING

Proactively search for attacker techniques and uncover hidden threats.



7. INCIDENT RESPONSE

Use MITRE to contain, eradicate and recover from incidents.



KEY THINGS TO REMEMBER



TACTIC = GOAL

Why the attacker is performing the action. (The end objective)



TECHNIQUE = ACTION

How the attacker achieves the goal. (The method used)



SUB-TECHNIQUE = SPECIFIC METHOD

More detailed way the technique is implemented.



MITRE IS NOT A TOOL

It is a knowledge base and a common language for defenders.



USE IT EVERYDAY

Map alerts, create detections, hunt threats, and improve response.

“

The more you understand attacker behavior, the better you can defend your organization.

– Blueteamers

MITRE IN ACTION (AT A GLANCE)



ALERT RECEIVED

SIEM / EDR / Firewall triggers an alert.



MAP TO MITRE

Identify the technique using MITRE ATT&CK.



INVESTIGATE

Understand the technique, check related behaviors.



HUNT & DETECT

Hunt for the same technique across systems.



RESPOND & IMPROVE

Take action, update detections and improve security.

WHERE TO FIND MITRE ATT&CK DATA?

- Official Website: attack.mitre.org
- GitHub: github.com/mitre/attack-stix-data
- ATT&CK Navigator for mapping & visualizations
- MITRE DEFEND for detections



PRO TIP FOR SOC ANALYSTS

- ✓ Use MITRE IDs in reports and tickets.
- ✓ Build detections based on techniques.
- ✓ Hunt using techniques, not just IOCs.
- ✓ Share a common language with your team.
- ✓ Stay updated with new techniques.



REMEMBER: MITRE ATT&CK is not a framework to memorize. It is a framework to understand attacker behavior, improve detections, investigate incidents and communicate effectively.

